Course Policy – Section – 005 **(MTWR)**

Math 223 – Calculus III – Spring 2024

CMNS 305 MTWR 3-3:50 PM

Instructor and Contact Information

Instructor name: Gregory Johnson and Laura Miller
Email address: gregoryjohnson@math.arizona.edu, lauram9@math.arizona.edu
Office hours: Gregory Johnson: MATH317 Monday 4:15-5:15 PM, Tuesday 12:30-1:30 PM, and Thursday 11 AM-12 PM, MTL121 Wednesday 11am-12pm
Laura Miller: Tuesday 5-6 pm (zoom), BSW 227: Wednesday 1 -3 pm
Tutoring: MW 10am—4pm (MTL121), TTh 4—6pm (online, access via Teams).
Course Webpage: http://math223.math.arizona.edu

Laura Miller: https://sites.google.com/site/swimflypump/

Description of Course

Math 223 Vector Calculus (4 semester credit hours) The course covers differential and integral calculus of functions of several variables. Topics include vector valued and scalar functions, partial derivatives, directional derivatives, chain rule, local optimization, double and triple integrals, the line integral, Green's theorem, Stokes' theorem and the Divergence theorem. Examinations are proctored.

Course Prerequisites or Co-requisites

MATH 129 or Math 223 with C or better.

Course Format and Teaching Methods

This class is scheduled to be taught in the IN-PERSON modality.

Course Objectives

- Recognize and sketch surfaces in three-dimensional space.
- Understand the algebraic and geometric properties of vectors and vector functions in two and three dimensions.
- Compute dot products and cross products and interpret their geometric meaning.
- Compute partial derivatives of functions of several variables and explain their meaning.
- Compute directional derivatives and gradients of scalar functions and explain their meaning.
- Parameterize curves in 2- and 3-space.
- Set up and evaluate double and triple integrals using a variety of coordinate systems.
- Evaluate integrals of scalar or vector fields and explain their physical interpretation.
- Apply Fundamental theorem of line integrals, Green's theorem, Divergence Theorem, and Stokes' theorem as relevant.

Expected Learning Outcomes

Upon completion of this course, students should be able to:

- Perform vector operations, determine equations of lines and planes, parametrize curves.

- Graphically and analytically synthesize and apply multivariable and vector-valued functions and their derivatives.
- Synthesize the key concepts differential, integral and multivariable calculus.
- Evaluate double and triple integrals in rectangular and curvilinear coordinates; and calculate areas and volumes using multiple integrals.
- Use double, triple and line integrals in applications, including Green's Theorem, Stokes' Theorem, Divergence Theorem and Fundamental theorem of line integrals.

Class D2L site: This site will serve as the central hub for communication and materials for our course, including but not limited to the following:

- Class announcements.
- Course policy and course calendar.
- Grades.
- Links to course materials and relevant websites such as WebAssign, Gradescope etc.
- Links to the final exam information.

Course material and Required Technology
The textbook (*Multivariable Calculus*; Sixth Edition by Hughes-Hallett et al.; published by Wiley) and the online homework system (WebAssign) are being delivered digitally via D2L through the Pay One Price program. You have complimentary access to the materials through Jan 23, 2024. You **must** take action (even if you have not accessed the materials) to opt-out if you want a refund and choose to source the content independently. The deadline to opt-out is 9:00pm MST, Jan 23, 2024. Refunds are posted to your Bursar account within 7-10 days business days of opting out. Note that opting out will affect material for all your classes for the semester. Refer to the Pay One Price FAQs at https://shop.arizona.edu/payoneprice_student for additional information.

Other materials such as class notes, videos, worksheets may be posted on Calculus website, on D2L or linked from D2L.

Students will use technology to examine mathematical concepts graphically and numerically. A graphing calculator is an important tool, and any model is allowed on the **final exam** provided **it cannot receive a wireless signal**. Students are expected to have a working calculator for each exam.

For this class you will need frequent access to a device (computer, tablet or smart phone) that can access relevant websites, scan and upload written work in pdf format, download and view pdf documents and access videos posted in D2L.

Class Meetings and Staying Current: The class will meet during the scheduled class time from 3-3:50 PM on MTWR in room CMNS 305. Our meetings allow us to develop our understanding of the content material and help us to meet the course goals and objectives using a variety of instructional techniques. This may include lectures, discussion, group work and other interactive class activities. You may be required to do reading or video assignments outside class time.

Class attendance policy:

- Students are expected to attend class and to be active participants. When learning mathematics that is computationally and conceptually challenging, passively watching videos is not nearly as effective as actively participating in class.
- If you feel sick, follow the University protocols.
- Inform your instructors if you will be missing a class meeting or an assignment deadline. Ask your instructor about the material or activities you will be missing.
- Non-attendance does not guarantee an automatic extension of due date or rescheduling of examinations/assessments. Students with Dean's excuse for official university activities should seek arrangement to make up missed work during their absence.
- Please communicate and coordinate any request directly with your instructor immediately. Accommodation is at your instructors' discretion, subject to course policy.
- Students who miss more than one week of class should contact the Dean of Students Office DOS-deanofstudents@email.arizona.edu to share the documentation about the challenges you are facing.

Course Communications

Announcements and important course information may be sent out via official University email or through D2L announcements. It is the student's responsibility to keep informed of any announcements, syllabus adjustments or policy changes made during scheduled classes, by email, or through D2L. (Preferred mode of communication with your instructors is via email. Your email should include both of your instructors in the email. Your email should have the class section # in the subject line. Also, anything you refer to from the HW should be given in detail, not by referring to question 37 section 3 chapter 13.)

Makeup Policy for Students Who Register Late

Those students who register (late) after the first-class meeting may have a chance to make up the WebAssign homework. But you should contact the instructor immediately after you register for class to make up for missing assignments. If you contact the instructor late you will not be given a chance to make up the Webassign assignments. No make-ups for quizzes, written assignments and activities.

Administrative Drop

A student may be administratively dropped for missing the first 3 days of classes without informing the instructors. To ensure a good commitment towards the course leading to a successful outcome, a student may be administratively dropped for 8 or more absences, not necessarily consecutively plus a failing status.

Classroom Behavior Policy

To foster a positive learning environment, students and instructors have a shared responsibility. We want a safe, welcoming, and inclusive environment where all of us feel comfortable with each other and where we can challenge ourselves to succeed. To that end, our focus is on the tasks at hand and not on extraneous activities (e.g., texting, chatting, web surfing, doing homework, listening to music etc.). Ear buds or head phones are not allowed in the classroom. The use of devices is restricted to our tasks at hand. (This means you should not have homework, other class material, or anything else on your device except the notes.) If you are seen using your device for something other than the task at hand, you will be asked to turn your device off for the duration of the class period.

Students observed engaging in distracting activity will be asked to cease this behavior, and if escalated, may be asked to leave lecture, and may be reported to the Dean of Students.

Safety on Campus and in the Classroom

For a list of emergency procedures for all types of incidents, please visit the website of the Critical Incident Response Team (CIRT): https://cirt.arizona.edu/case-emergency/overview

Also watch the video available at https://arizona.sabacloud.com/Saba/Web_spf/NA7P1PRD161/common/learningeventdetail/crtfy000000000003560

Graded components:

- **WebAssign Homework:** (15%) An online homework system called WebAssign will be used for problems assigned from the text. The due dates for all assignments are posted in WebAssign. *It is your responsibility to keep track of assignment due dates.*
  If you miss a homework deadline, you can request an auto-extension within 24 hours of the original deadline using the *Request Extension* feature in WebAssign. (The button will appear immediately after the deadline has passed.) You will have up to 24 hours from your request time to complete the assignment. To encourage completing assignments on time, a 5% penalty will apply to the additional work completed on that assignment during the extension period. You can only request an extension once for each assignment.
  All assignments are equally weighted. The lowest 3 scores will be dropped. There are no make-up assignments beyond the auto-extension. Dean's excuse for official university business will be accommodated if requested.
- **Quizzes, Written homework assignments, and Class activities: (**10%)
  - **Quizzes:** (5%) Short, **timed/proctored** quizzes will be given regularly. You will be taking the quizzes via Gradescope or in-person in class. Each quiz will be timed (usually around 25 minutes) and is worth 10 points. All quizzes are open book and open notes. (We recommend that you attempt the quizzes without using notes or your book because you will not be allowed to use any of that during exams.) Quizzes are administered on the dates listed on the calendar. Each quiz will be equally weighted and two will be dropped at the end of the semester. There will be no make-up quizzes. Dean's excuse for official university business will be accommodated if requested.
  - **Written assignments:** (5%) *Hand-written* homework that requires demonstrating reasonings and calculations using proper notation will be given. These problems will come from the text and/or from a set of problems created by your instructor. Each assignment will be equally weighted and three will be dropped at the end of the semester. There will be no make-up written assignments. Dean's excuse for official university business will be accommodated if requested.
  - **In-class activities:** (0%)
    - In-class activities consist of working on problems in class and participating in peer discussions on assigned work. **Passively attending class alone does not earn credits.**
    - The purpose of this component is to encourage student engagement, improve attention/focus and to verify understanding.
  - Any assignments/work you upload will be submitted on Gradescope. If you do not select the page the problem is on you will receive zero points for the problem. You need to have one problem per page/picture. **Do not have more than one problem on a page/picture. If multiple problems are on the same page, you will receive zero points for the problems.** Also, all your assignments should be handwritten clearly with all the steps shown clearly and each step should

follow from the preceding steps. Assignments should not be typed. Any work you submit in-person in class and if you use multiple pages, it should be stapled. Otherwise only the first page gets graded.

- **FTC Mini Test:** (5%) This mandatory mini test evaluates all fundamental theorems that will be given in class, and will not be dropped. There will be no make-up for the FTC Mini Test. Dean's excuse for official university business will be accommodated if requested.
- **Midterms:** (45%) There are three (50 minute) in-class midterm exams, administered and proctored in-person. The tentative dates for these exams are included in the course calendar. All exams are equally weighted. Exams are closed references and no electronic devices are allowed except for an approved calculator.

  Missed and Replacement Exam Policy: If a student misses a midterm exam and does not have a Dean's excuse for official university activities, that student will receive a score of 0 on the missed exam. If a student must miss an exam and has an official Dean's excuse for university activities, that student must contact the instructors at least one week prior to the exam. A make-up exam must be completed before the official exam time unless the Dean's excuse extends to at least two days before the official exam time, in which case the make-up exam must be completed on the next business day after the student's return. Otherwise, the student will receive a grade of 0 on the missed exam.

  A student's lowest midterm score will be replaced by their final exam percentage if their final exam percentage is higher than their lowest midterm score. No more than one midterm score can be replaced, even if a student's two lowest midterm scores are identical.

  According to university policy, no exams will be held during the week of April 29th.

  Final Examination: (25%) The proctored, in person final exam is a 110-minute comprehensive common exam that is closed book/notes. It is scheduled for Tuesday, May 7th from 1-3pm (see the University's Final Exam Schedule at http://www.registrar.arizona.edu/schedules/finals.htm). The only allowed electronic device is a graphing calculator that cannot receive wireless signals. *Do not plan or schedule any activities during the scheduled final exam time.* The location of the common Final exam will be announced toward the last week of class.

Additional information and a study guide can be found at https://math223.math.arizona.edu.

The University's Exam regulations will be strictly followed https://www.registrar.arizona.edu/courses/final-examination-regulations-and-information.

Grading Scale and Policies

Midterms (3)	WebAssign	Quizzes/ Written assignments/ Participation	FTC Mini Test	Final Exam	Total
45%	15%	10%	5%	25%	100%

Your final course grade will be *no lower than* the following:

A	B	C	D	E
100-90%	89.99-80%	79.99-70%	69.99-60%	59.99-0%

No extra credit or bonus points are offered on this course.

Requests for incomplete (I) or withdrawal (W): Requests must be made in accordance with University policies (ref: http://catalog.arizona.edu/policy/grades-and-grading-system#incomplete and

http://catalog.arizona.edu/policy/grades-and-grading-system#Withdrawal)

The dates are listed on Dates & Deadlines | Office of the Registrar (arizona.edu). A brief excerpt is included in the class calendar on this document.

Dispute of Grade Policy: Any questions regarding the grading of any assignment, quiz, or exam need to be cleared up within one week after the graded item has been returned.

Use of Generative AI policy: In this course any and all uses of generative artificial intelligence (AI)/large language model tools such as ChatGPT, Dall-e, Google Bard, Microsoft Bing, etc. will be considered a violation of the Code of Academic Integrity, specifically the prohibition against submitting work that is not your own. This applies to all assessments in the course, including written assignments, quizzes, exams, and class activities. The following actions are prohibited:

- entering all or any part of an assignment statement or test questions as part of a prompt to a large language model AI tool;
- incorporating any part of an AI-written response in an assignment; and
- submitting your own work for this class to a large language model AI tool for iteration or improvement.

University-wide Policies link

The Links to the following UA policies are provided here, https://academicaffairs.arizona.edu/syllabus-policies:

• Absence and Class Participation Policies

• Threatening Behavior Policy

• Accessibility and Accommodations Policy

• Code of Academic Integrity

• Nondiscrimination and Anti-Harassment Policy

Additional Resources for Students

- **UA Academic policies and procedures** are available at http://catalog.arizona.edu/policies
- **Student Assistance and Advocacy information** is available at http://deanofstudents.arizona.edu/student-assistance/students/student-assistance
- **Academic advising**: If you have questions about your academic progress this semester, or your chosen degree program, please note that advisors at the Advising Resource Center can guide you toward university resources to help you succeed. For assistance email to advising@arizona.edu
- **Life challenges**: If you are experiencing unexpected barriers to your success in your courses, please note the Dean of Students Office is a central support resource for all students and may be helpful. The Dean of Students Office can be reached at 520-621-2057 or DOS-deanofstudents@email.arizona.edu.
- **Physical and mental-health challenges**: If you are facing physical or mental health challenges this semester, please note that Campus Health provides quality medical and mental health care. For medical appointments, call (520-621-9202. For After Hours care, call (520) 570-7898. For the Counseling & Psych Services (CAPS) 24/7 hotline, call (520) 621-3334.

Confidentiality of Student Records
http://www.registrar.arizona.edu/personal-information/family-educational-rights-and-privacy-act-1974-ferpa?topic=ferpa

Accessibility and Accommodations: At the University of Arizona, we strive to make learning experiences as accessible as possible. If you anticipate or experience barriers based on disability or pregnancy, please contact the Disability Resource Center (520-621-3268, https://drc.arizona.edu) to establish reasonable accommodations.

Tentative Daily Scheduled Topics/Activities- Spring 2024

Month	*Date*	*Day*	*Section*	*Topics*	*Assessments*	*Platform used*
Jan	10	**W**	13.1	Intro, 13.1-Displacement Vectors		
	11	**R**	13.1	13.1-Displacement Vectors		
	15	**M**		***Martin Luther King Jr Holiday***		
	16	**T**	13.2,13.3	13.2-Vectors in General, 13.3-The Dot Product		
	17	**W**	13.3	13.3-The Dot Product	***Last day to add/change sections usingUaccess***	
	18	**R**		Catch-up/Problem Session	**Written HW 1**	gradescope
					Quiz 1 (Friday 1/19)	gradescope
	22	**M**	13.4	13.4- The cross Product		
	23	**T**	12.1	12.1- Functions of two variables	***Last day to drop without W***	
	24	**W**	12.2	12.2-Graphs and Surfaces		
	25	**R**		Catch-up/Problem Session	**Written HW 2**	gradescope
					Quiz 2 (Friday 1/26)	gradescope
	29	**M**	12.3	12.3-Contour Diagrams		
	30	**T**	12.4	12.4-Linear Functions		
	31	**W**	12.5	12.5-Functions of Three Variables		
Feb	1	**R**		Catch-up/Problem Session	**Written HW 3**	gradescope
					Quiz 3 (Friday 2/2)	gradescope
	5	**M**	14.1	14.1-The Partial Derivative		
	6	**T**	14.2	14.2-Computing Partial Derivatives Analytically		
	7	**W**		Review/ Catch-up		
	8	**R**		**Test 1**	**Test 1**	In class

	12	**M**	14.3	14.3-Local Linearity & Differentials		
	13	**T**	14.4	14.4-Gradients & Directional Derivatives in the Plane		
	14	**W**	14.5	14.5- Gradients & Directional Derivatives in space		
	15	**R**		Catch-up/Problem Session	**Written HW 4**	gradescope
					Quiz 4 (2/16)	gradescope
	19	**M**	14.6	14.6-The Chain Rule		
	20	**T**	14.7	14.7-Second-Order Partial Derivatives		
	21	**W**	16.1	16.1-The Definite Integral of a Function of Two Variables		
	22	**R**		Catch-up/Problem Session	**Written HW 5**	gradescope
					Quiz 5 (2/23)	gradescope
	26	**M**	16.2	16.2-Iterated Integrals		
	27	**T**	16.4	16.4-Double Integrals in Polar Coordinates		
	28	**W**	16.3	16.3-Triple Integrals		
	29	**R**		Catch-up/Problem Session	**Written HW 6**	gradescope
March	**4-10**			***Spring break***	**Quiz 6 (3/1)**	gradescope
	11	**M**	16.5	16.5-Integrals in Cylindrical & Spherical Coordinates		
	12	**T**	17.1	17.1-Parametric Curves		
	13	**W**		Review/ Catch up		
	14	**R**		**Test 2**	**Test 2**	In class
	18	**M**	17.1	17.1-Parametric Curves		
	19	**T**	17.2	17.2-Motion, Velocity, &Acceleration		
	20	**W**	17.3	17.3-Vector Fields		
	21	**R**	18.1	18.1-The Idea of a Line Integral	**Written HW 7**	gradescope
					Quiz 7 (3/22)	gradescope
	25	**M**	18.1	18.1-The Idea of a Line Integral		
	26	**T**	18.2	18.2-Computing Integrals over Parameterized Curves	***Last Day to withdraw Last day to GRO***	
	27	**W**	18.3	18.3-Gradient Fields & Path Independent Fields		

	28	**R**		Catch-up/Problem Session	**Written HW 8**	gradescope
					Quiz 8 (3/29)	gradescope
April	1	**M**	18.4	18.4-Path Dependent Vector Fields & Green's Theorem		
	2	**T**	18.4	18.4-Path Dependent Vector Fields & Green's Theorem		
	3	**W**	19.1	19.1-The Idea of Flux Integrals		
	4	**R**		Catch-up/Problem Session	**Written HW 9**	gradescope
					Quiz 9 (4/5)	gradescope
	8	**M**	19.1, 19.2	19.1-The Idea of Flux Integrals, 19.2 Flux Integrals for Cylinders, & Spheres		
	9	**T**	19.2	19.2 Flux Integrals for Graph.		
	10	**W**		Review/ Catch-up		
	11	**R**		**Test 3**	**Test 3**	In class
	15	**M**	19.3	19.3-The Divergence of a Vector Field		
	16	**T**	19.4	19.4 The Divergence Theorem		
	17	**W**	19.4	19.4 The Divergence Theorem		
	18	**R**		Catch-up/Problem Session	**Written HW10**	gradescope
					Quiz 10 (4/19)	gradescope
	22	**M**	20.1	20.1-The Curl of a Vector Field		
	23	**T**	20.2	20.2-Stokes' Theorem		
	24	**W**	20.2	20.2-Stokes' Theorem		
	25	**R**		**FTC mini-test**	**FTC mini-test**	In class
					Written HW11 (4/26)	gradescope
	29	**M**	15.1	15.1-Critical Points: Local Extrema & Saddle Points		
	30	**T**		Review		
May	1	**W**		Review- ***Last Day of class***		
	2	**R**		***Reading day-No Class***		
	7	**T**		**Final Exam- 110 minutes (comprehensive-Chapters 12-20)**	**Final Exam (1-3pm)**	In Person –Gradescope grading

Subject to Change Statement
Information contained in the course syllabus, other than the grade and absence policy, may be subject to change with advance notice, as deemed appropriate by the instructor.

Where to go, who to call if you're in crisis:

Located in Tucson? Call the Community-Wide Crisis Line 24 hours a day, 7 days a week at 520-622-6000.

Are you a University of Arizona student? If it is not an emergency and you are a UA student, call or walk-in to Counseling and Psych Services at 520-621-3334 Monday - Friday.

Walk-in triage is available between 9 am and 4 pm Monday - Friday.

Are you a concerned friend? Concerned friends can find out more about helping a friend who might be experiencing problems through our Friend 2 Friend website.

Resources for sexual assault, relationship violence, and stalking.

24-Hour Hotlines:

The National Suicide Prevention Lifeline is a 24-hour, toll-free, confidential suicide prevention hotline available to anyone in suicidal crisis or emotional distress. By dialing 1-800-273-TALK (8255), the call is routed to the nearest crisis center in our national network of more than 150 crisis centers. The Lifeline's national network of local crisis centers provides crisis counseling and mental health referrals day and night.

Crisis Text Line: Text HOME to 741741 from anywhere in the United States, anytime, about any type of crisis. A live, trained Crisis Counselor receives the text and responds, all from a secure online platform. Find out more about how it works at crisistextline.org.

Suicide Prevention for LGBTQ Youth through the Trevor Project:

• The Trevor Lifeline is a 24/7 suicide hotline: 866-4-U-TREVOR (1-866-488-7386)

• **TrevorChat:** Online instant messaging available 7 days a week, 3 pm - 10 pm ET (12 pm -- 7 pm PT)

• **TrevorText:** Confidential and secure resource that provides live help for LGBTQ youth with a trained specialist, over text messages. Text TREVOR to 1-202-304-1200 (available 7 days a week, 3 pm - 10 pm ET, 12 pm -- 7 pm PT)

Veterans' Suicide Prevention Lifeline: 1-800-273-TALK (1-800-273-8255)

SAMHSA Treatment Referral Hotline (Substance Abuse): 1-800-662-HELP

(1-800-662-4357)

National Sexual Assault Hotline: 1-800-656-HOPE (1-800-656-4673)

Loveisrespect (National Dating Abuse Helpline): Call 1-866-331-9474 (TTY: 1-866-331-8453). Text LOVEIS to 22522 - you'll receive a response from a peer advocate prompting you for your question. Go ahead and text your comment or question and we will reply.